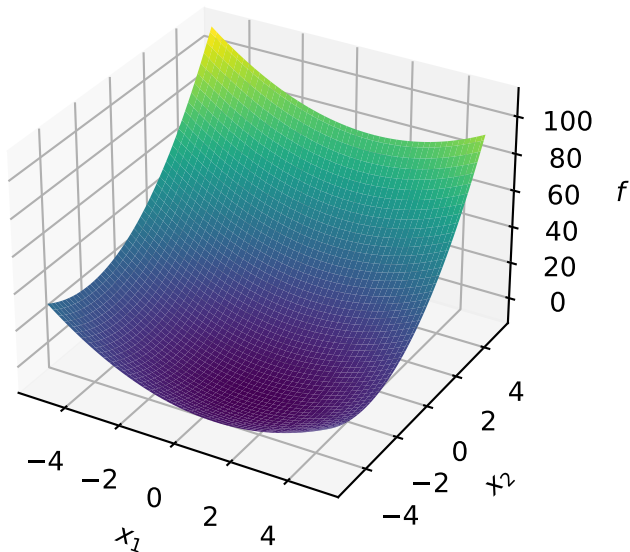
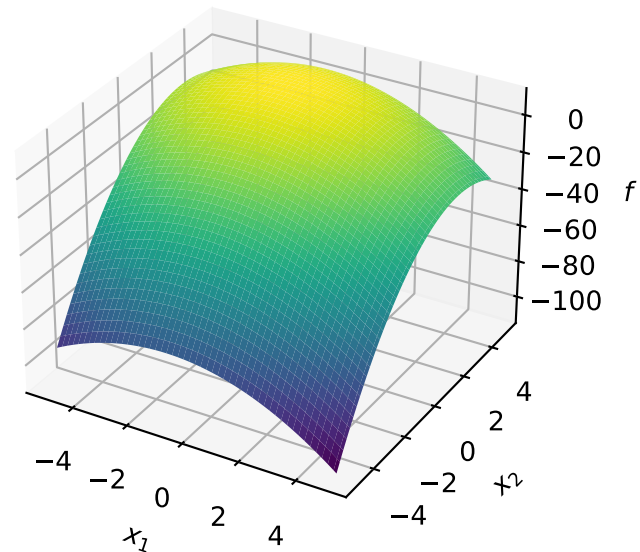


aim: solve $\mathbf{Ax} = \mathbf{b}$, here $\mathbf{x} \in \mathbb{R}^2$
 Quadratic forms $f = \frac{1}{2}\mathbf{x}^T\mathbf{Ax} - \mathbf{b}^T\mathbf{x}$

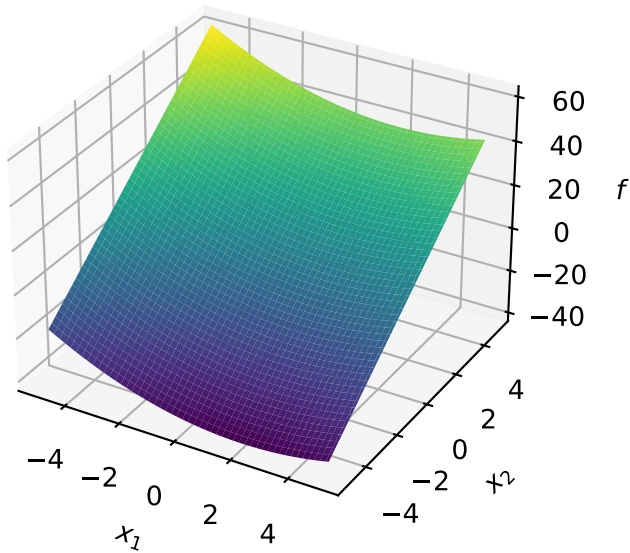
A is positive definite
 solution at minimum



A negative definite
 solution at maximum



A is singular (non-invertible)
 no unique solution



A is non-singular and indefinite
 solution at saddle point

